

Problem Decomposition: Goal Tree and AO*

What is Problem Decomposition?

Problem decomposition means breaking a complex problem into smaller, easier-to-solve subproblems.

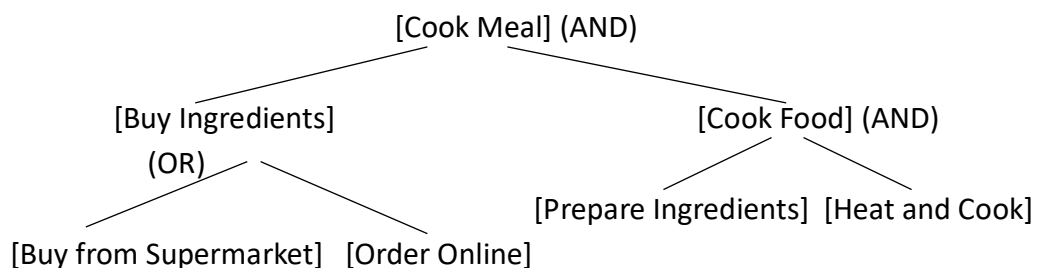
This approach makes it easier to solve large problems by focusing on individual parts one at a time.

1. Goal Tree (And-Or Tree)

Definition:

A Goal Tree is a tree-like structure that represents a main goal and the subgoals needed to achieve it. It contains:

- AND nodes – all subgoals must be completed.
- OR nodes – completing any one subgoal is enough.



Explanation:

- “Cook Meal” is an AND node: you must do both Buy Ingredients and Cook Food.
- “Buy Ingredients” is an OR node: you can buy from supermarket OR order online.
- “Cook Food” is an AND node: both Prepare Ingredients and Heat and Cook must be done.

2. AO* Algorithm (And-Or Search Algorithm)

Definition:

The AO* (And-Or Star) algorithm is a heuristic search algorithm used to find the best solution path in goal trees or And-Or graphs.

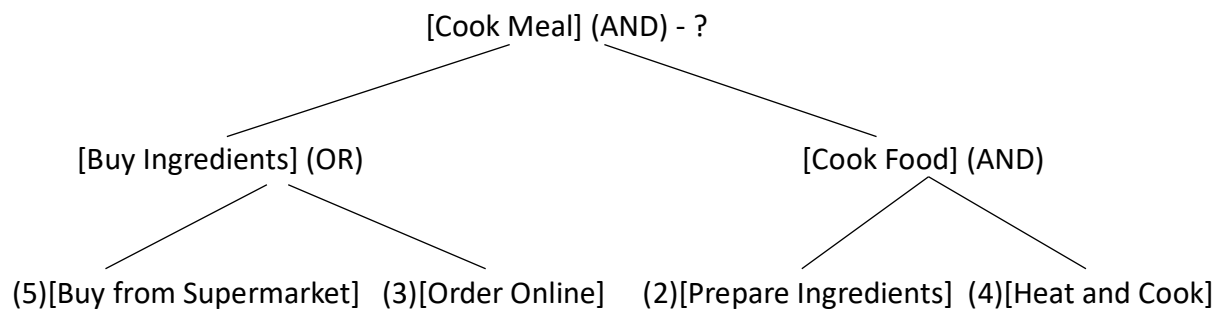
Features:

- Handles both AND and OR relationships.
- Uses heuristic values (estimated costs) to guide search.
- Finds the optimal solution graph, not just a path.

Steps of AO* (Simplified):

1. Start from the main goal (root).
2. Expand nodes based on AND/OR.
3. Use heuristics to choose the most promising path.
4. Mark solved subproblems.
5. Continue until an optimal solution graph is found.

Example



Explanation:

Costs:

- Buy from Supermarket = 5
- Order Online = 3
- Prepare Ingredients = 2
- Heat and Cook = 4

AO* Chooses:

- Order Online (3) instead of Supermarket (5) — lower cost
- Prepare Ingredients (2) and Heat and Cook (4) are both required

Total Cost: $3 + 2 + 4 = 9$

So AO* will choose:

- Order Online
- Prepare Ingredients
- Heat and Cook

Because it is the cheapest valid path to solve the goal.

Summary:

Concept	Goal Tree	AO* Algorithm
Purpose	Shows structure of subgoals	Finds best solution in a Goal Tree
Node Types	AND, OR	AND, OR
Strategy	Visual decomposition	Heuristic-based search
Output	Tree of possible paths	Best solution subgraph

Note:

- Goal Trees help you understand the structure of problems.
- AO* helps you find the best way to solve them, especially when there are many choices with different costs.